

100%

Excellent!

Name: _____

These questions assess your understanding of the material from Chapter(s) 20, 21 of OpenStax College Physics. Please write your name on the sheets of paper that you use. Carefully read each question. Credit is only awarded if you answer correctly and clearly show all major steps necessary to find your answer; credit is not awarded for missing work, unclear work, or the use of memorized equations absent from the equation sheet. Half credit may be awarded for insufficient work that suggests some degree of understanding. Half credit is awarded for correct numerical answers with incorrect or missing units. Correct numerical answers do not guarantee credit.

1. What is the value of the peak voltage for 150-V AC power?

$$V_{RMS} = \frac{V_0}{\sqrt{2}} \rightarrow V_0 = \sqrt{2} V_{RMS} = \sqrt{2} (150 \text{ V}) = \boxed{212.132 \text{ V}}$$

ANSWER: 212.132 V ✓

2. A 1.6-m-long copper wire of diameter 1.9 mm is connected to a 20.-V battery at 6.6°C. Calculate the percent change in the power delivered by the wire when it is heated to 89.°C. The resistivity of copper at 6.6°C is $1.7 \times 10^{-8} \Omega \cdot \text{m}$ and its linear thermal expansion coefficient is $3.9 \times 10^{-3} \frac{1}{^\circ\text{C}}$.

$$6.6^\circ\text{C}: P = \frac{V^2}{R} = \frac{V^2}{\frac{\rho L}{A}} = \frac{V^2 \pi r^2}{\rho L} = \frac{V^2 \pi \frac{d^2}{4}}{\rho L} = \frac{V^2 \pi d^2}{4 \rho L}$$

ANSWER: -24.320% ✓

$$P = \frac{(20 \text{ V})^2 \pi (0.0019 \text{ m})^2}{4 (1.7 \times 10^{-8} \frac{\text{V} \cdot \text{m}}{\text{A}}) (1.6 \text{ m})} = 41695.403 \text{ W}$$

$$89^\circ\text{C}: P = \frac{V^2 \pi d^2}{4 L \rho_0 (1 + \alpha \Delta T)} = \frac{(20 \text{ V})^2 \pi (0.0019 \text{ m})^2}{4 (1.6 \text{ m}) (1.7 \times 10^{-8} \frac{\text{V} \cdot \text{m}}{\text{A}}) (1 + (3.9 \times 10^{-3} \frac{1}{^\circ\text{C}}) (89^\circ\text{C} - 6.6^\circ\text{C}))} = 31554.915 \text{ W}$$

$$\% \text{ change} = \frac{31554.915 \text{ W} - 41695.403 \text{ W}}{41695.403 \text{ W}} \times 100\% = \boxed{-24.320\%}$$

3. Suppose the voltage supplied to a circuit is 4.3 V. There are three resistors wired in series: $R_1 = 2.3 \Omega$, $R_2 = 2.1 \Omega$, and $R_3 = 8.3 \Omega$. Calculate the current in the circuit.

$$V = IR \rightarrow I = \frac{V}{R} = \frac{V}{R_s} = \frac{4.3 \text{ V}}{2.3 \frac{\text{V}}{\text{A}} + 2.1 \frac{\text{V}}{\text{A}} + 8.3 \frac{\text{V}}{\text{A}}} = \boxed{0.339 \text{ A}}$$

ANSWER: 0.339 A ✓

+3

- ✓ 4. What current flows through a high-voltage transmission line to transmit 150 MW of power at 140 kV?

$$P = IV \rightarrow I = \frac{P}{V} = \frac{150 \times 10^6 \text{ A} \cdot \text{V}}{140 \times 10^3 \text{ V}} = \boxed{1071.429 \text{ A}}$$

ANSWER: 1071.429 A

- ✓ 5. Calculate the power dissipated by a car headlight that draws a current of 3.3 A when it has a voltage applied to it of 13. V.

$$P = IV = (3.3 \text{ A})(13 \text{ V}) = \boxed{42.9 \text{ W}}$$

ANSWER: 42.9 W

- ✓ 6. Suppose a 0.017-mA current flows through a handheld calculator. How long does it take a charge of $1.9 \times 10^{-5} \text{ C}$ to flow through the calculator?

$$I = \frac{\Delta Q}{\Delta t} \rightarrow \Delta t = \frac{\Delta Q}{I} = \frac{1.9 \times 10^{-5} \text{ C}}{0.017 \times 10^{-3} \frac{\text{C}}{\text{s}}} = \boxed{1.118 \text{ s}}$$

ANSWER: 1.118 s

7. A 11.2-V car battery supplies a starter motor drawing 140 A for 2.3 s. Calculate the electrical energy delivered by the battery during this time.

$$IV = \frac{E}{t} \rightarrow E = tIV = (2.3 \text{ s})(140 \text{ A})(11.2 \text{ V})$$

$$E = \boxed{3606.4 \text{ J}}$$

ANSWER: 3606.4 J ✓

8. An incandescent lightbulb has a 1.9-cm-long filament and operates at 116 V to deliver 73 W. Calculate the diameter of the bulb's filament, assuming it to be a cylindrical tungsten wire, given that the resistivity of tungsten is $5.6 \times 10^{-8} \Omega \cdot \text{m}$.

$$P = \frac{V^2}{R} \rightarrow R = \frac{V^2}{P} = \frac{\rho L}{A} = \frac{\rho L}{\pi r^2} = \frac{\rho L}{\pi d^2} = \frac{4\rho L}{\pi d^2}$$

ANSWER: $2.711 \times 10^{-6} \text{ m}$ ✓

$$\pi d^2 = \frac{4\rho L}{\frac{V^2}{P}} = \frac{4\rho L P}{V^2} \rightarrow d = \sqrt{\frac{4\rho L P}{V^2 \pi}} = \sqrt{\frac{4(5.6 \times 10^{-8} \frac{\text{V} \cdot \text{m}}{\text{A}})(0.019 \text{ m})(73 \text{ A} \cdot \text{V})}{(116 \text{ V})^2 \pi}}$$

$$d = \boxed{2.711 \times 10^{-6} \text{ m}}$$

9. Suppose the voltage supplied to a circuit is 4.2 V. There are three resistors wired in parallel: $R_1 = 1.0 \Omega$, $R_2 = 4.8 \Omega$, and $R_3 = 1.8 \Omega$. Calculate the power output of the battery.

$$P = \frac{V^2}{R} = \frac{V^2}{R_p} = \frac{(4.2 \text{ V})^2}{(1^{-1} + 4.8^{-1} + 1.8^{-1})^{-1} \frac{\text{V}}{\text{A}}} = \boxed{31.115 \text{ W}}$$

ANSWER: 31.115 W ✓

+3

10. A 71-m-long copper wire must carry 13 A without dissipating more than 540 W as heat. Calculate the wire's minimum diameter given that copper's resistivity is $1.7 \times 10^{-8} \Omega \cdot m$.

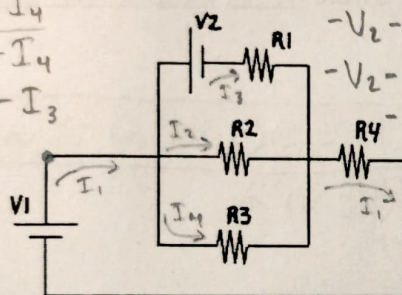
$$P = I^2 R \rightarrow R = \frac{P}{I^2} = \frac{\rho L}{A} = \frac{4\rho L}{\pi d^2} \rightarrow \pi d^2 = \frac{4\rho L}{\frac{P}{I^2}} = \frac{4\rho L I^2}{P}$$

$$d = \sqrt{\frac{4\rho L I^2}{P\pi}} = \sqrt{\frac{4(1.7 \times 10^{-8} \frac{V \cdot m}{A})(71 m)(13 A)^2}{(540 A \cdot V)\pi}} = 6.935 \times 10^{-4} m$$

11. Solve for the current that passes through resistor R2 given $V1 = 10 V$, $V2 = 9 V$, $R1 = 3 \Omega$, $R2 = 2 \Omega$, $R3 = 8 \Omega$, $R4 = 1 \Omega$.

ANSWER: $3.319 A$

$$\begin{aligned} I_1 &= I_2 + I_3 + I_4 \\ I_3 &= I_1 - I_2 - I_4 \\ I_4 &= I_1 - I_2 - I_3 \end{aligned}$$



$$-V_2 - I_3 R_1 - I_1 R_4 + V_1 = 0$$

$$-V_2 - (I_1 - I_2 - I_4) R_1 - I_1 R_4 + V_1 = 0$$

$$-9 - 3I_1 + 3I_2 + 3I_4 - I_1 + 10 = 0$$

$$-4I_1 + 3I_2 + 3I_4 = -1$$

$$-I_4 R_3 - I_1 R_4 + V_1 = 0$$

$$-8I_4 - I_1 + 10 = 0$$

$$-I_1 - 8I_4 = -10$$

$$\frac{-35}{8} I_1 + 3I_2 = \frac{-19}{4}$$

$$-I_1 - 2I_2 = -10 \rightarrow I_1 = -2I_2 + 10$$

$$\frac{-35}{8} (-2I_2 + 10) + 3I_2 = \frac{-19}{4}$$

$$I_2 = 3.319 A$$

$$I_1 = 3.362 A$$

$$I_4 = 0.830 A$$

$$I_3 = -0.787 A$$

$$-I_2 R_2 - I_1 R_4 + V_1 = 0 \quad I_4 = \frac{-I_1 + 10}{8}$$

$$-2I_2 - I_1 + 10 = 0$$

$$-I_1 - 2I_2 = -10$$

$$-4I_1 + 3I_2 + 3\left(\frac{-I_1 + 10}{8}\right) = -1$$

$$-4I_1 + 3I_2 - \frac{3}{8} I_1 + \frac{15}{4} = -1$$

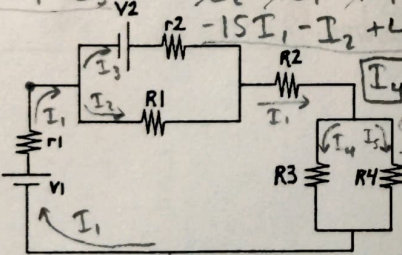
12. Solve for the current that passes through resistor R3 given $V1 = 3 V$, $V2 = 4 V$, $r1 = 7 \Omega$, $r2 = 7 \Omega$, $R1 = 1 \Omega$, $R2 = 4 \Omega$, $R3 = 2 \Omega$, $R4 = 4 \Omega$.

ANSWER: $0.177 A$

$$I_1 = I_2 + I_3 = I_4 + I_5 \quad -I_2 - 4I_1 - 4I_1 + 4I_4 + 3 - 7I_1 = 0$$

$$I_5 = I_1 - I_4$$

$$I_3 = I_1 - I_2$$



$$-15I_1 - I_2 + 4I_4 = -3$$

$$I_4 = 0.177 A$$

$$-11I_1 - I_2 - 2I_4 = -3$$

$$15I_1 + I_2 - 4I_4 = 3$$

$$4I_1 - 6I_4 = 0$$

$$-18I_1 + 7I_2 - 2I_4 = -7$$

$$-105I_1 - 7I_2 + 28I_4 = -21$$

$$-123I_1 + 26I_4 = -28$$

$$I_1 = \frac{6I_4}{4}$$

$$-123\left(\frac{6I_4}{4}\right) + 26I_4 = -28$$

$$I_4 = 0.177 A$$

$$-I_2 R_1 - I_1 R_2 - I_4 R_3 + V_1 - I_1 r_1 = 0$$

$$-I_2 - 4I_1 - 2I_4 + 3 - 7I_1 = 0$$

$$-11I_1 - I_2 - 2I_4 = -3$$

$$V_2 - I_3 r_2 - I_1 R_2 - I_4 R_3 + V_1 - I_1 r_1 = 0$$

$$V_2 - (I_1 - I_2) r_2 - I_1 R_2 - I_4 R_3 + V_1 - I_1 r_1 = 0$$

$$4 - 7I_1 + 7I_2 - 4I_1 - 2I_4 + 3 - 7I_1 = 0$$

$$-18I_1 + 7I_2 - 2I_4 = -7$$

$$-I_2 R_1 - I_1 R_2 - I_5 R_4 + V_1 - I_1 r_1 = 0$$

$$-I_2 R_1 - I_1 R_2 - (I_1 - I_4) R_4 + V_1 - I_1 r_1 = 0$$

PHY202 Exam 3 on chapter(s): 20, 21

1. $V_0 = 212. \text{ V} \therefore V_{RMS} = \frac{V_0}{\sqrt{2}}$
(Chapter 20, Section (5) Alternating Current v. Direct Current - cp1e_ch20_ex9)
2. $\% \Delta P = -24.3\% \therefore \rho(T) = \rho_0(1 + \alpha \Delta T), R = \frac{\rho L}{A}, P = \frac{V^2}{R}$
(Chapter 20, Section (4) Electric Power and Energy - tkd3)
3. $I = 0.339 \text{ A} \therefore R_s = R_1 + R_2 + R_3, V = IR$
(Chapter 21, Section (1) Resistors in Series and Parallel - cp1e_ch21_ex1)
4. $I_{RMS} = 1.07 \times 10^3 \text{ A} \therefore P_{ave} = I_{RMS} V_{RMS}$
(Chapter 20, Section (5) Alternating Current v. Direct Current - cp1e_ch20_ex10)
5. $P = 42.9 \text{ W} \therefore P = IV$
(Chapter 20, Section (4) Electric Power and Energy - cp1e_ch20_ex7)
6. $t = 1.12 \text{ s} \therefore I = \frac{\Delta Q}{\Delta t}$
(Chapter 20, Section (1) Current - cp1e_ch20_ex1)
7. $E = 3.61 \times 10^3 \text{ J} \therefore P = IV = \frac{E}{t}$
(Chapter 20, Section (4) Electric Power and Energy - tkd1)
8. $d = 2.71 \times 10^{-6} \text{ m} \therefore P = \frac{V^2}{R}, R = \frac{\rho L}{A} \text{ and } A = \pi r^2$
(Chapter 20, Section (4) Electric Power and Energy - tkd2)
9. $P = 31.1 \text{ W} \therefore \frac{1}{R_p} = \frac{1}{R_1} + \dots + \frac{1}{R_n}, V = IR, P = IV$
(Chapter 21, Section (1) Resistors in Series and Parallel - cp1e_ch21_ex2)
10. $d_{min} = 6.94 \times 10^{-4} \text{ m} \therefore R = \frac{\rho L}{A}, P_{int} = I^2 R$
(Chapter 20, Section (4) Electric Power and Energy - tkd4)
11. $I = 3.32 \text{ A} \therefore \text{Kirchhoff's rules}$
(Chapter 21, Section (3) Kirchhoff's Rules - tkd_circ1)
12. $I = 0.177 \text{ A} \therefore \text{Kirchhoff's rules}$
(Chapter 21, Section (3) Kirchhoff's Rules - tkd_circ2)